

HERNANDO, PANTINE B. CHALLENGE - 12 - CRYPTOGRAPHY

BSIT-3

morse-code 



Medium Cryptography picoCTF 2022 morse_code

AUTHOR: WILL HONG

Description

Morse code is well known. Can you decrypt this?

Download the file [here](#).

Wrap your answer with picoCTF{}, put underscores in place of pauses, and use all lowercase.

Hints ?

1

21,723 users solved



75% Liked



 picoCTF{FLAG}

Submit Flag

Challenge Overview:

The challenge provides an audio file (typically a `.wav` or `.mp3`) containing Morse code beeps. To solve it, you need to decode the audio, format it according to specific rules, and wrap it in the flag format.

1. Analysis

The description gives us three critical formatting rules for the final flag:

1. **Wrap** the answer with `picoCTF{}`.
2. **Replace pauses** (gaps between words) with underscores `_`.
3. Use **all lowercase** letters.

2. The Solution Path

Method A: Automated Decoding (Easiest)

Since human ears can be fallible (and slow), the most efficient way to solve this is using an online spectrum analyzer or Morse decoder.

1. **Download** the audio file from the challenge link.
2. **Upload** it into an online tool like Data Border <https://databorder.com/transfer/morse-sound-receiver/> The tool will process the audio beeps and output the text automatically.

 DATA BORDER

Morse Code Sound & Vibration Listener

This is javascript only morse code listener. Use it with something that emits Morse code as sound or vibrations

Audio Input

Microphone:

Pre-Recorded Audio File File: "morse_chal.wav"

Decoder Settings & Info...

Minimum volume

Maximum volume

Volume threshold

WPM ☐ Manual

Farnsworth WPM

Frequency (Hz) ☐ Manual

Received Data

3. Formatting the Flag

Let's say the decoded Morse code results in: **WH47 H47H G0D WR0U6H7**

Following the challenge instructions:

- **Lowercase:** **wh47 h47h g0d wr0u6h7**
- **Underscores for pauses:** **wh47_h47h_g0d_wr0u6h7**
- **Wrap in picoCTF{}**: **picoCTF{wh47_h47h_g0d_wr0u6h7}**

Received Data

WH47 H47H 90D W20U9H7

4. Key Takeaways

- **Signal Processing:** In CTFs, audio "cryptography" is often just data encoded in a different physical medium (sound).
- **Instruction Adherence:** Most "incorrect" submissions on this challenge come from forgetting the lowercase rule or using spaces instead of underscores.